

An integral tensor outside the algebraic completion

Candidate counterexample for arXiv:2404.16338

19 July 2026

Status: candidate counterexample, likely valid. The integral-representation space of Definition 4.1 need not be the completion of the algebraic tensor product for its defining seminorms. The counterexample is explicit and uses two copies of a step-function model of ℓ_∞ .

1 Source and question

The source is Eva-Maria Hekkelman, Edward McDonald, and Teun D. H. van Nuland, *Multiple operator integrals, pseudodifferential calculus, and asymptotic expansions*, arXiv:2404.16338, 2024 [1]. In Definition 4.1 they introduce $\Gamma_0 \boxtimes_i \cdots \boxtimes_i \Gamma_n$: its elements are functions admitting representations

$$\phi(\lambda_0, \dots, \lambda_n) = \int_{\Omega} \prod_{j=0}^n a_j(\lambda_j, \omega) d\nu(\omega) \quad (1)$$

with finite product-seminorm integral. The corresponding seminorm is the infimum of that integral over all representations. After noting the inclusion of the algebraic tensor product, the authors ask whether the displayed space is, in general, its completion for these seminorms.

2 Proof intuition

The integral permits a continuum of rank-one factors without requiring the factor-valued maps to be norm measurable. Independent signs exploit exactly that freedom: their correlation is the infinite diagonal matrix. By contrast, an algebraic tensor samples to a finite-rank matrix. The seminorm in the definition dominates pointwise uniform norm, and the infinite diagonal stays at least $1/2$ away in uniform norm from every finite-rank matrix. Thus it cannot lie in the algebraic completion for the very seminorm used to define the integral space.

3 The counterexample

For $m \geq 1$, let

$$I_m = [m, m+1), \quad t_m = m + \frac{1}{2}.$$

Let Γ be the space of all bounded measurable real-valued functions on $\mathbb{R}$ which vanish outside $[1, \infty)$ and are constant on each I_m . Equip Γ with

$$\|f\|_{\Gamma, k} := \|f\|_{\infty} \quad (k \in \mathbb{N}).$$

Thus Γ is a Banach function space isometric to ℓ_∞ , and all its listed seminorms coincide.

Namely, given the second representation, a representation of the first type can be obtained by putting [ST19, p.48]

$$a_j(\lambda_j, \sigma) := \frac{b_j(\lambda_j, \sigma)}{\|b_j(\cdot, \sigma)\|_{L_\infty^\beta(E_j)}}, \quad \nu(\sigma) := \|b_0(\cdot, \sigma)\|_{L_\infty^\beta(E_0)} \cdots \|b_n(\cdot, \sigma)\|_{L_\infty^\beta(E_n)} \mu(\sigma).$$

Mainly for notational purposes, we introduce the following definition. It is inspired by the integral projective tensor product appearing in [Pel06], a precise study can be found in [Nik23].

Definition 4.1. Let $\Gamma_0, \dots, \Gamma_n$ be function spaces of bounded measurable functions $\mathbb{R} \rightarrow \mathbb{R}$ equipped with (semi)norms $\|\cdot\|_{\Gamma_{i,k}}$. We define $\Gamma_0 \boxtimes_i \cdots \boxtimes_i \Gamma_n$ as the set of functions $\phi : \mathbb{R}^{n+1} \rightarrow \mathbb{C}$ for which there exists a decomposition

$$\phi(\lambda_0, \dots, \lambda_n) = \int_\Omega a_0(\lambda_0, \omega) \cdots a_n(\lambda_n, \omega) d\nu(\omega) \quad (10)$$

where (Ω, ν) is a σ -finite measure space, $a_i : \mathbb{R} \times \Omega \rightarrow \mathbb{C}$ is measurable, $a_i(\cdot, \omega) \in \Gamma_i$, the functions $\omega \mapsto \|a_i(\cdot, \omega)\|_{\Gamma_{i,k}}$ are measurable for each i and k , and

$$\int_\Omega \|a_0(\cdot, \omega)\|_{\Gamma_{0,k_0}} \cdots \|a_n(\cdot, \omega)\|_{\Gamma_{n,k_n}} d|\nu|(\omega) < \infty. \quad (11)$$

We define the seminorm

$$\|\phi\|_{\Gamma_0 \boxtimes_i \cdots \boxtimes_i \Gamma_n, k_0, \dots, k_n}$$

to be the infimum of the quantity (11) over all representations (10).

Remark 4.2. We have that

$$\Gamma_0 \otimes \cdots \otimes \Gamma_n \subseteq \Gamma_0 \boxtimes_i \cdots \boxtimes_i \Gamma_n,$$

where $\otimes$ denotes the algebraic tensor product for topological vector spaces.

Note that due to Lemma 2.5 if $a : \mathbb{R} \times \Omega \rightarrow \mathbb{C}$ is measurable and $a(\cdot, \omega) \in S^\beta(\mathbb{R})$, we have that $\omega \mapsto \|a(\cdot, \omega)\|_{S^\beta(\mathbb{R}), k}$ is measurable, and the same claim holds for $T^\beta(\mathbb{R})$ and $L_\infty^\beta(E)$. Hence the construction in Definition 4.1 can be applied without this extra assumption.

For $L_\infty^\beta(E)$ with E a spectral measure, this gives the integral projective tensor product

$$L_\infty^{\beta_0}(E_0) \boxtimes_i \cdots \boxtimes_i L_\infty^{\beta_n}(E_n) = L_\infty^{\beta_0}(E_0) \hat{\otimes}_i \cdots \hat{\otimes}_i L_\infty^{\beta_n}(E_n),$$

which appears in particular for $\beta_0 = \cdots = \beta_n = 0$ in the works by Peller [Pel06, pp.6, 7].

We refrain from answering the question whether the space in Definition 4.1 is in general the completion of the algebraic tensor product under the given seminorms.

Figure 1: Definition 4.1, its seminorm, the algebraic inclusion, and the completion question on page 22 of the source paper.

Define

$$\phi(x, y) := \sum_{m=1}^{\infty} \mathbf{1}_{I_m}(x) \mathbf{1}_{I_m}(y). \quad (2)$$

The sum is pointwise either zero or one.

Theorem 1. The function ϕ belongs to $\Gamma \boxtimes_i \Gamma$, but for every $\psi \in \Gamma \otimes \Gamma$ and every pair of seminorm indices k_0, k_1 ,

$$\|\phi - \psi\|_{\Gamma \boxtimes_i \Gamma, k_0, k_1} \geq \frac{1}{2}.$$

Consequently, $\Gamma \boxtimes_i \Gamma$ is not the completion of $\Gamma \otimes \Gamma$ under the given seminorms. In particular, the general completion question after Definition 4.1 of the source paper has a negative answer.

Proof. Let $\Omega = \{-1, 1\}^{\mathbb{N}}$ with its product sigma algebra and product probability measure $\mathbb{P}$, and

write $\varepsilon_m(\omega) = \omega_m$ for its coordinate functions. Put

$$a(x, \omega) := \sum_{m=1}^{\infty} \varepsilon_m(\omega) \mathbf{1}_{I_m}(x). \quad (3)$$

The map $(x, \omega) \mapsto a(x, \omega)$ is jointly measurable: it is defined on a countable measurable partition by measurable coordinate functions. For every ω , the function $a(\cdot, \omega)$ lies in Γ and has all seminorms equal to one. Independence and centering of the signs give

$$\int_{\Omega} a(x, \omega) a(y, \omega) d\mathbb{P}(\omega) = \sum_{m, l \geq 1} \mathbf{1}_{I_m}(x) \mathbf{1}_{I_l}(y) \mathbb{E}(\varepsilon_m \varepsilon_l) = \phi(x, y).$$

Only one summand in each spatial variable can be nonzero, so the displayed calculation is pointwise finite. This is a representation of the form (1), of cost one for every pair (k_0, k_1) . Hence $\phi \in \Gamma \boxtimes_i \Gamma$.

We next record that the defining integral seminorm dominates uniform norm. Indeed, for every representation

$$h(x, y) = \int_{\Sigma} b(x, \sigma) c(y, \sigma) d\mu(\sigma)$$

with factors in Γ , one has

$$|h(x, y)| \leq \int_{\Sigma} \|b(\cdot, \sigma)\|_{\infty} \|c(\cdot, \sigma)\|_{\infty} d|\mu|(\sigma).$$

Taking first the supremum over (x, y) and then the infimum over representations yields

$$\|h\|_{\infty} \leq \|h\|_{\Gamma \boxtimes_i \Gamma, k_0, k_1}. \quad (4)$$

Now take an algebraic tensor

$$\psi(x, y) = \sum_{s=1}^r f_s(x) g_s(y)$$

and sample it at the points t_m . Let

$$A_{mn} := \psi(t_m, t_n), \quad r_m := (A_{mn})_{n \geq 1} \in \ell_{\infty}.$$

All rows r_m lie in the finite-dimensional subspace

$$F = \text{span}\{(g_s(t_n))_{n \geq 1} : 1 \leq s \leq r\} \subseteq \ell_{\infty}.$$

Suppose, toward a contradiction, that

$$\epsilon := \sup_{m, n} |A_{mn} - \delta_{mn}| < \frac{1}{2}.$$

Then $\|r_m\|_{\infty} \leq 1 + \epsilon$ for every m . A bounded subset of the finite-dimensional normed space F is totally bounded. Thus there are distinct p, q such that

$$\|r_p - r_q\|_{\infty} < 1 - 2\epsilon.$$

Writing $e_m = (\delta_{mn})_{n \geq 1}$ for the m th identity row gives

$$1 = \|e_p - e_q\|_{\infty} \leq \|e_p - r_p\|_{\infty} + \|r_p - r_q\|_{\infty} + \|r_q - e_q\|_{\infty} < 2\epsilon + (1 - 2\epsilon) = 1,$$

a contradiction. Therefore

$$\sup_{m,n} |\psi(t_m, t_n) - \delta_{mn}| \geq \frac{1}{2}. \quad (5)$$

Because $\phi(t_m, t_n) = \delta_{mn}$, equations (4) and (5) prove

$$\|\phi - \psi\|_{\Gamma \boxtimes_i \Gamma, k_0, k_1} \geq \|\phi - \psi\|_\infty \geq \frac{1}{2}.$$

Thus no net or sequence of algebraic tensors can converge to ϕ for the given seminorms. $\square$

4 Verification, scope, and novelty

The construction checks every condition in Definition 4.1 directly: the parameter measure is finite, the factor is jointly measurable, every section belongs to Γ , the norm map is the constant one, and the product-norm integral equals one. The separation uses the exact seminorm from that definition, not the larger Banach projective norm. Since that seminorm dominates uniform norm here, it is definite and no quotient-completion issue arises.

This fully answers the words “in general” in the negative. It does not decide density for the special spaces $S^\beta(\mathbb{R})$, $T^\beta(\mathbb{R})$, or the spectral L_∞^β spaces used elsewhere in the source.

Before promotion, the run indexes and ledgers were searched using the source arXiv id and the core phrases “integral projective tensor”, “algebraic tensor completion”, “Rademacher”, and “diagonal”. A bounded web/arXiv search used the exact question sentence, the source title and id, and close integral-projective-tensor phrases. No separate answer was found. Moreover, Hekkelman’s 2026 thesis repeats the question verbatim in Remark 3.2.2 [2]. This is strong recent status evidence, not an exhaustive novelty guarantee.

Human review is recommended. The highest-value checks are the interpretation of completion in the source sentence and the total-boundedness separation of the sampled rows.

References

- [1] E.-M. Hekkelman, E. McDonald, and T. D. H. van Nuland, *Multiple operator integrals, pseudodifferential calculus, and asymptotic expansions*, arXiv:2404.16338, 2024. <https://arxiv.org/abs/2404.16338>.
- [2] E.-M. Hekkelman, *Trace Formulas*, doctoral thesis, 2026, Chapter 3, Remark 3.2.2. <https://www.emhekkelman.nl/uploads/phdthesis.pdf>.